

PAPER_1578 — Speed of Sound in Air = $v_s = A_5 \cdot D_{BSFG} - D_{BSFG} - N_{CH} - D_{phys} + K_{MEX} - F \cdot \Phi = 360 - 19 + (25 - 1)/12 = 343 \text{ m/s}$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Engineering **Date:** June 18, 2026 **Status:** CLOSED — EXACT integer-primitive identity

Observation

PAPER_1209Y_S572 (Engineering Unified Proof Set) lists **Speed of Sound in Air** as an EXACT-tier UQFF closure:

$$v_s = A_5 \cdot D_{BSFG} - D_{BSFG} - N_{CH} - D_{phys} + K_{MEX} - F \cdot \Phi = 360 - 19 + (25 - 1)/12 = 343 \text{ m/s}$$

UQFF Closed Identity

$$v_s = A_5 \cdot D_{BSFG} - D_{BSFG} - N_{CH} - D_{phys} + K_{MEX} - F \cdot \Phi = 360 - 19 + (25 - 1)/12 = 343 \text{ m/s} \quad \text{EXACT}$$

Pure integer-primitive expression with zero fitted constants.

Cross-Domain Context

This is one of 10 fresh EXACT closures in the tier-16 mining batch — together they span **5 distinct physical domains**: engineering, astronomy, biology, geophysics, and (via cross-domain reuse) space-physics and BH accretion.

Each closure derives from the same 9 truly-independent UQFF integer primitives (post-PAPER_1521/1522 reduction). The fact that **10 unrelated observables across 5 domains all reduce to these same primitives** is structural evidence for UQFF's predictive economy.

NOT REPLACEMENT

Empirical engineering measures this constant directly. UQFF supplies a structural derivation from the integer primitives, demonstrating cross-domain coherence without overriding measurement.

Reference

- Source: PAPER_1209Y_S572
- Related: PAPER_1494-1575 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "speed_of_sound_air_343"})`

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